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ORIGINAL ARTICLE

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Population dynamical consequences of reduced predator switching at low total prey densities

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Abstract Several models of rapid switching by a predator in a two-prey environment are analyzed. The goal is to determine how the dynamics of the system and the potential indirect effects between prey are affected by the dependence of switching on total prey density. In exploring this question, the difference between the population-level consequences of switching in stable and cycling predator-prey systems is also examined. We concentrate on reduced switching at low densities, a feature that is likely because of the difficulty of distinguishing between two very low densities. The main findings are: (1) switching in unstable systems can produce positive indirect effects of one prey species on the other; and (2) reduced switching at low densities can greatly alter the dynamics of the system and the indirect effects between prey. Both of the possibilities are only evident in cycling systems. Reduced switching at low total prey densities leads to heavier predation on the slower-growing prey when both prey species are rare. As a consequence, there is a lag in the recovery of the slower-growing prey species after predator densities fall, and the dynamics of the two prey become desynchronized. The net result is increased indirect interactions between prey, and a greater likelihood of exclusion of the slower growing prey. The analysis of these models suggests a need for more empirical work to determine whether switching is reduced by very low total prey densities, and to study the long-term dynamics that occur in systems with switching predators.

Keywords Apparent competition · Chaos · Choice behavior · Predation · Population cycles · Switching

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Introduction

Predator switching occurs when a predator's ratio of consumption rates of two prey types increases as the ratio of their densities increases (Murdoch 1969). Switching is thus equivalent to positively frequency dependent predation (Gendron 1987). Switching is likely when two or more prey species are nutritionally substitutable, but the predator cannot maximize its consumption of all prey types simultaneously (Murdoch and Oaten 1975; Oaten and Murdoch 1975; Abrams 1987). Switching has been observed in a wide range of consumer species (Lawton et al. 1974; Murdoch and Oaten 1975; Cornell 1976, Bergelson 1985; Joern 1988; Hughes and Croy 1993; Wickham 1995). It has been proposed to be a factor stabilizing predator-prey dynamics generally (Murdoch and Oaten 1975; Post et al. 2000), and specifically in systems where voles are one of the main prey species (Hanski et al. 2001). However, switching does not always promote stability, as shown by recent models analyzed by Bernstein et al. (1999), Abrams (1999, 2000), Abrams and Kawecki (1999) and van Baalen et al. (2001). Switching can promote the coexistence of many competing prey species (Roughgarden and Feldman 1974; Teramoto et al. 1979), and often decreases the strength of apparent competition between non-competing prey (Holt 1977, 1984; Holt and Kotler 1987; Abrams and Matsuda 1996). Thus, switching can potentially be a very potent mechanism for increasing prey species diversity. Switching also has implications for natural selection on predator avoidance mechanisms in prey, such as predator swamping (Ims 1990).

In spite of its importance, empirical work has ignored many aspects of switching. For most systems, switching has been studied by varying the ratio of densities of two prey species while presenting the predator with a constant total number of prey individuals (Murdoch 1969; Lawton et al. 1974; Bergelson 1985; Joern 1988; Hughes and Croy 1993). However, there are few, if any, studies that determine whether the frequency dependence

revealed by this type of experiment depends on the total prey density. Similarly, the majority of previous theoretical studies of switching (Tansky 1978; Teramoto et al. 1979; Matsuda 1985; Abrams 1987; Matsuda et al. 1987; van Baalen and Sabelis 1999; Post et al. 2000; van Baalen et al. 2001) have followed Murdoch's original definition, and assumed that the current ratio of prey densities is the sole determinant of predator preference. This assumption makes switching independent of total prey densities. A number of other formulations were used in earlier models (reviewed in Holt 1983), but none of these discussed the consequences of density-dependence for the dynamics of systems with switching.

It is reasonable to expect that, at low total densities, stochastic factors will make estimation of prey densities difficult, and could obscure or eliminate frequency dependence. We refer to this as 'reduced switching' at low prey densities. Such changes in switching, dependent on density, would have the greatest impact on systems that undergo repeated changes in total prey density, which is the case for any system in which there are predator-prey cycles, or in which there are seasonal declines that affect all prey species. Here, we show that reduced switching at low densities does in fact cause significant differences in the ability of prey species to coexist, and in the indirect effects between them when they do coexist. These theoretical results call for more empirical work, both to study the dependence of switching on total prey density, and to determine the population dynamics of systems in which both switching and predator-prey cycles occur.

Another aspect of switching that is often ignored is the time lag required by the predator population to switch, following a change in the relative abundances of prey. This has received limited empirical (Bergelson 1985; Hughes and Croy 1993) and theoretical (Abrams 1999, 2000) attention, and will be the subject of a subsequent article (Abrams and Matsuda, unpublished).

Models and analysis

We consider a predator species that searches for two prey species within a single habitat where they are homogeneously distributed. The models explored below undergo limit cycles for some parameters, due to a saturating predator functional response: specifically the two-prey version of Holling's (1959) disk equation. It has long been known (Rosenzweig and MacArthur 1963) that such responses can generate cycles when predators are efficient or prey density dependence is sufficiently weak. The two prey species are assumed to be nutritionally substitutable for the predator, and to have equal ratios of energy content (B) to handling time (h). However, the prey may differ in their absolute values of these parameters. This assumption is appropriate when digestion time is the main component of handling time, because both digestion time and energy content are expected to be proportional to prey size. Ma et al. (2003) analyze some similar models in which prey differ in their profitabilities.

The reason why switching is advantageous over separate specialist types is that prey densities fluctuate in time; endogenously driven cycles are one of the primary reasons for such fluctuations (Ellner and Turchin 1995; Kendall et al. 1998). However, fluctuations alone do not guarantee that switching is advantageous relative to having a mix of specialist predator types, each of which consumes only one of the prey types. Here, we assume that switching is advantageous because switching generalist predators achieve a higher consumption rate than a mixture of specialists when both prey are abundant, as a result of a convex tradeoff between the predator's attack rates on the two prey. Switching generalists have greater fitness than do fixed generalists because of their ability to more effectively capture whichever prey species is currently most common. Switching can also be favored in cycling systems without a convex tradeoff, because switching predators can achieve a higher mean intake rate than can any specialist when predators have a convex numerical response. This second mechanism will be treated elsewhere.

The tradeoff between the predator's two attack rates, c_1 and c_2 (for prey species 1 and 2, respectively) is determined by a continuous variable, z, which ranges between 0 and 1, and reflects the degree of specialization on prey 1. The tradeoff relationship is given by:

$$c_1 = C_1 z^{\omega}, \text{ and } c_2 = C_2(1-z)^{\omega} \tag{1}$$

where C_i reflects the sensitivity of capture rate to the trait z. The exponent ω is a positive constant that determines the shape of the tradeoff between the two capture rates; $\omega > 1$ implies a concave relationship, where generalists have lower capture rates than the average of the two specialist types. Here we assume $\omega < 1$, which implies a convex relationship with an advantage for generalists. When ω is closer to 0, the tradeoff becomes more nonlinear, the relative advantage of a generalist becomes greater, and the magnitude of switching is reduced. The value of z is assumed to change adaptively with prey densities in one of two possible ways described later.

We investigate a population dynamics model in which component functions are both mathematically simple and widely used. Thus, the two prey species are assumed to have logistic growth with the possibility of migration from an external source. The predator's functional response for a given z is a two-species version of Holling's (1959) disk equation, and the predator's numerical response is either linear or is itself described by a disk equation. These assumptions result in the following model for the population dynamics of the two species:

$$\begin{aligned}
\frac{dR_1}{dt} &= i_1 + r_1 R_1\left(1-\frac{R_1}{K_1}\right) - \frac{c_1(z)R_1 N}{1+hc_1(z)R_1+ehc_2(z)R_2} \\
\frac{dR_2}{dt} &= i_2 + r_2 R_2\left(1-\frac{R_2}{K_2}\right) - \frac{c_2(z)R_2 N}{1+hc_1(z)R_1+ehc_2(z)R_2} \\
\frac{dN}{dt} &= N\left(\frac{B(c_1(z)R_1+ec_2(z)R_2)}{1+hc_1(z)R_1+ehc_2(z)R_2} - D\right)
\end{aligned} \tag{2}$$

where the parameters are:

- r_j the intrinsic growth rate of prey j
- K_j the carrying capacity of prey j
- i_j the immigration of prey j into the system
- c_j the per capita capture rate of prey j by a searching predator individual
- h the handling time for an item of prey 1
- B the energy (or limiting nutrient) content of an individual of prey 1
- e the proportional difference in both energy content and handling time of prey 2 relative to prey 1 (implied by our assumption of equal prey profitability)
- D the per capita death rate of the predator

Adaptive switching implies that z changes with prey densities in a manner that tends to maximize the predator's food intake rate. This maximization depends on the relationship between c_1 and c_2. If both prey have positive densities, the assumption that $\omega < 1$ means that there is an intermediate optimal value of z at any point in time. This value can be determined by differentiating the expression for intake rate with respect to z (assuming the tradeoff given by Eq. 1), setting this to zero, and solving for the optimal z, yielding,

$$z_{\text{opt}} = \frac{(C_1R_1)^u}{(eC_2R_2)^u + (C_1R_1)^u} \tag{3}$$

where $u = 1/(1-\omega)$. Note that this expression means that z is purely a function of the ratio of prey densities. Given our assumptions, this optimum is independent of the other parameters of the functional or numerical responses. If $\omega \geq 1$, the optimum z is 0 when $C_1R_1 < C_2R_2$, and 1 when this inequality is reversed. However, here we only consider the case of $\omega < 1$. Equations 2a, b, and c incorporate switching by substituting Eq. 3 for z, given that the attack rates c_1 and c_2 are described by Eq. 1.

It is not necessary that switching be based on exact maximization of fitness, as was assumed above. In fact, it seems improbable that an animal could detect small absolute differences in prey densities as easily as it detects large absolute differences, given identical ratios. An alternative formulation that depends on the absolute difference between intake rates, and therefore avoids this problem, is:

$$z_{\text{opt}} = \frac{1}{1 + \exp\left[\eta(eC_2R_2 - C_1R_1)\right]} \ . \tag{4}$$

Here, the difference in prey densities (weighted by capturability and relative energy content) determines diet choice; either equal intake rates from both prey, or small but unequal intake rates from both, results in a z_{opt} close to 1/2, implying a highly generalized diet. The parameter η determines how rapidly z approaches 0 or 1 as the difference in intake rates increases: large values of η make the switching function closer to a step-function of relative resource densities. Thus, a large η could make Eq. 4 approximate Eq. 3 closely for a range of very small total prey densities. However, it would then not be a good approximation when total prey density was larger. Another difference between the two expressions is that Eq. 4 predicts a significant decrease in the predator's attack rate on prey 1 as that prey becomes rarer, even when prey 2 is at negligible densities. There is very little change in the attack rates according to Eq. 3 under these circumstances. However, it is again not clear which formula is more appropriate here. A low density of prey 1 could cause the predator to periodically search for prey 2, even when the latter is at near-zero density, thus reducing consumption of prey 1.

Another way to make switching depend on absolute densities is to modify Eq. 3 by adding a small positive quantity, δ, to each of the terms in that expression:

$$z_{\text{opt}} = \frac{(\delta + C_1R_1)^u}{(\delta + eC_2R_2)^u + (\delta + C_1R_1)^u} \ . \tag{5}$$

The general effect of δ is to reduce the degree of predator preference. This parameter can be thought of as reflecting errors in the predator's estimation of prey densities. If δ is small relative to both C_iR_i, then it has little effect on the predicted predator preference. However, if δ is large relative to both C_iR_i, then Eq. 5 is close to 1/2, and the diet is predicted to be relatively generalized. Unlike Eq. 4, but like Eq. 3, Eq. 5 predicts that the consumption of a particular prey is approximately independent of its own density when the other prey is at negligible density. When δ is nonzero in Eq. 5 or η is nonzero in Eq. 4, there is a nonzero minimum attack rate by the predator on each prey, given the tradeoff in Eq. 1. As pointed out by Comins and Hassell (1976), coexistence of two prey is no longer guaranteed in this case (as it is under Eqs. 1 and 3).

We are interested in how the effects of switching on prey densities and dynamics differ depending on the choice of Eq. 3, 4, or 5 to represent the predator's diet choice. Because cycling seemed likely to increase the difference between models with purely frequency-dependent switching (Eq. 3) and those with density- and frequency-dependent switching (Eqs. 4, 5), we make this comparison of alternative models for both stable and cycling systems.

The comparison of Eqs. 3 and 4 requires that we choose parameters ω $[=(u-1)/u]$ and η that give comparable consumption rates across a range of relative prey densities for an appropriate total prey density. Here, we arbitrarily assume that Eq. 3 is the 'true' relationship, and we adjust η to obtain the best fit to that relationship: this depends on the total prey density. The most common choice for that total density in empirical work has been ambient prey density, either in the presence or absence of the predator. In both cases, we use the ambient density in a single prey system with prey species 1. That density is K_1 (in most of our models $K_1 = K_2$) in the absence of the predator, and

approximately $D/[C_1(B-Dh)]$ in the presence of the predator. We choose a value of η by minimizing the absolute difference between Eqs. 3 and 4, averaged over the range of prey densities along the line $R_2+R_1=K$ or, alternatively, the line $R_2+R_1=D/[C_1(B-Dh)]$. These ranges correspond to sets of empirical measurements of prey consumption over a range of different relative densities of the two prey, for a given total prey density. The two alternative total prey densities yield two values of η. We then numerically integrate Eqs. 2a, b, and c using Eq. 4 assuming each of the two alternative values for η, and compare the dynamics and mean densities in these two cases with those predicted by Eq. 3. For systems that have stable equilibria, we can determine the densities by numerically solving the equations that specify the equilibria.

For a given total prey density, fitting the parameter η in the manner described above results in an excellent fit of Eq. 4 to Eq. 3. Figure 1 is an example showing both formulas for the case where total prey density is K ($K_1=K_2=1$), and where $\omega=0.8$. In this case, the maximum difference in the preference parameter z predicted by the two formulas is 0.00017, which means that the line for the best fit of Eq. 4 almost exactly overlaps the line specified by Eq. 3. Figure 1 also shows the shape of the switching curve predicted by Eq. 4 for values of η that are five times larger or smaller than the best-fit value. The value of η obtained by fitting Eq. 4 to data based on Eq. 3 depends on the total prey density used. If the data is based on a total prey density of K/x, where $x>1$, the best fit value for η will be x multiplied by the best fit value obtained with a total prey density of K. The difference between Eq. 5 and Eq. 3 increases with the magnitude of δ and, for a given δ, is largest when one or both prey densities are small.

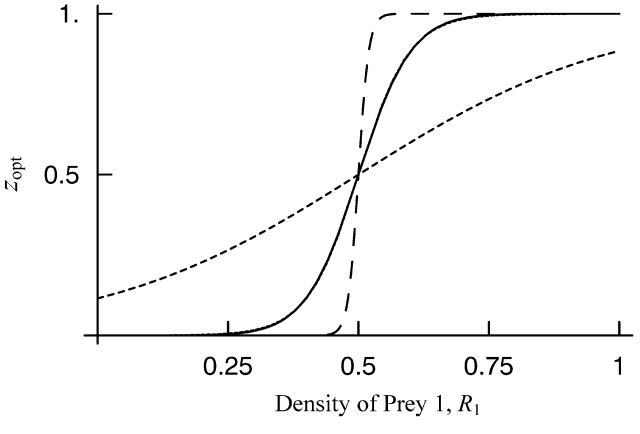


Fig. 1 The preference parameter, z, predicted by Eq. 3 (*solid line*) and three versions of Eq. 4 (*dashed lines*), for a case with attack rates (C_i) of 5, and a tradeoff parameter ω of 0.8. Total prey density is 1. The *very short dashed line* that is nearly coincident with the *solid line* resulted from fitting the parameter η to minimize the difference between the curves for Eqs. 3 and 4. The other two *dashed lines* assume values of η five times larger (*long-dashed line*) or smaller (*short-dashed line*) than the best-fit value of $\eta=2.0384$

The indirect interactions between prey species in stable systems

The consequences of switching in stable systems have been analyzed in previous work (Holt 1983, 1984; Abrams 1987; Abrams and Matsuda 1996), so will be treated relatively briefly here. If the tradeoff described by Eq. 1 is linear ($\omega=1$), then there is no effect of introducing either prey species on the equilibrium density of the other prey when it occurs together with the predator. The increase in predator density as a result of the second prey is exactly cancelled by the fact that the predator splits its foraging effort between the two prey species. For the nonlinear tradeoff considered here ($\omega<1$), each prey species is reduced in density when the second prey species is added or, if the second species is present, when the growth rate of that species is increased. However, switching reduces the impact that would be expected if the attack rates of the predator had fixed intermediate values.

One way to quantify the indirect effects between the two prey species is to examine how altering the intrinsic growth rate of one prey changes the population size of the other prey, when both occur together with the predator. This relationship illustrates the indirect effect of prey 2 on prey 1, and it allows calculation of this indirect effect for a range of different initial growth rates of species 2, and for a variety of magnitudes of perturbation in that growth rate. The arithmetic mean population densities of the two prey also provide a basis for quantitatively comparing the dynamics produced by different switching functions. Figure 2 is the first of several figures that show mean densities of both prey across a range of intrinsic growth rates of prey 2. Figure 2 uses Eq. 3 for z_{opt}, and examines the consequences of changing the nonlinearity of the tradeoff in attack rates. Like subsequent figures which have the same format, Fig. 2 assumes that prey 1 has an intrinsic growth rate $r_1=1$. Each panel in Fig. 2 plots the density of both prey as a function of r_2 for values from 0 to 2. Thus, the density of prey species 1 when it is alone with the predator is its density when $r_2=0$. The density of prey 1 when it occurs together with a second prey species can be read from the graph, for a range of potential values of r_2. For example, if the two prey are initially equivalent, $r_2=1$, and the consequence of removing prey 2 for prey 1 can be determined by comparing the density of prey 1 at $r_2=1$ and at $r_2=0$. Figure 2 also makes it possible to determine the consequences of increasing or decreasing the intrinsic growth rate of species 2 on the density of species 1 by simply comparing R_1 for the original and the altered values of r_2. Carrying out this comparison using different pairs of r_2 values can be used to see how the sign and magnitude of the indirect effect via the predator depends on the magnitude of the perturbation to the per capita growth rate of the focal prey (species 2).

In Fig. 2, the handling times for both species have been set to zero; this ensures that the equilibrium point with both species present will be stable. The figure shows that the indirect interaction between the two prey

becomes weaker when the tradeoff becomes more linear (ω closer to 1): this is reflected in the lower slope of the plot of the density of prey 1 versus the intrinsic growth rate of prey 2 in Fig. 2B than in Fig. 2A. Figure 2B also shows that prey 2 can sustain a dramatic decline in growth rate with very little change in either its density or that of the other prey when ω is close to 1. Furthermore, the effect of decreased r_2 on R_1 is small over nearly the entire range of possible values of r_2. Tradeoffs which are more nonlinear (e.g., $\omega = 1/2$; Fig. 2A) result in less pronounced switching and greater apparent competition. However, in both panels of Fig. 2 the interaction between prey species is always negative, regardless of the magnitude of the change in intrinsic growth rate that is used to measure the interaction or the initial asymmetry in growth rates of the two species. The interaction may be positive when the predator has direct density dependent effects on its own growth rate (Abrams and Matsuda 1996).

Figure 3 compares the indirect effects between prey species predicted by Eqs. 3 and 4 when systems are stable and there is a moderately nonlinear tradeoff given by

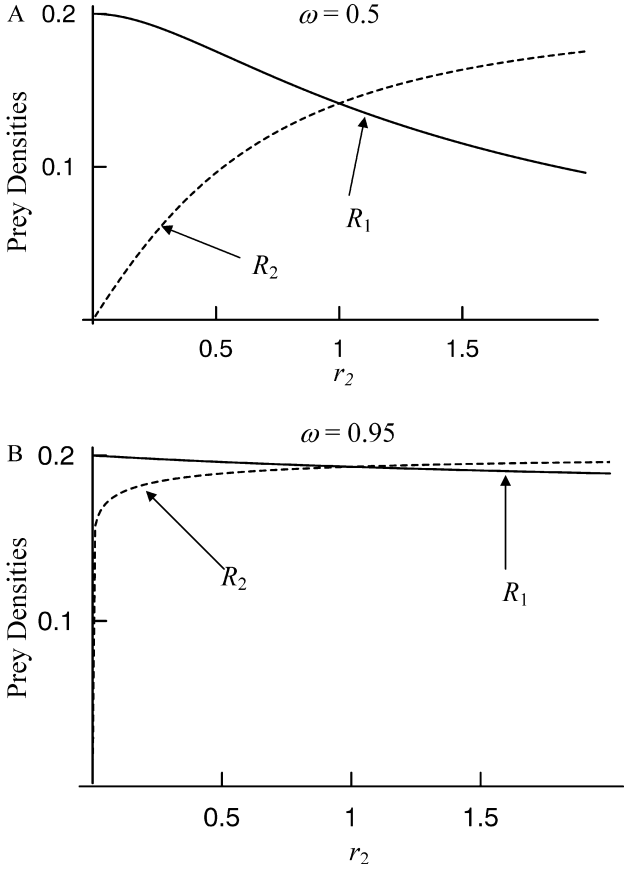


Fig. 2A, B Densities of the two prey as a function of the intrinsic growth rate of prey 2 in a stable system where switching is based on Eq. 3. The *solid line* is the density of prey 1 and the *dashed line* is the density of prey 2. **A** assumes a strongly convex tradeoff with $\omega = 0.5$, while **B** assumes a nearly linear tradeoff with $\omega = 0.95$. The remaining parameters for both panels are: $C_i = 5$; $e = 1$; $i_i = 0$; $r_1 = 1$; $K_i = 1$; $B = 0.1$; $h = 0$; $D = 0.1$

$\omega = 0.8$. In Fig. 3A–C, handling time (h) is zero; it is positive in Fig. 3D. Figure 3A displays prey densities as a function of r_2 with switching based on Eq. 3. Figure 3B assumes Eq. 4, with the value of η fitted using a total prey density of 1 ($= K$). The difference in the predicted prey population sizes for the two switching formulas comes about because Eq. 4 predicts a more generalized diet when total prey density is low. There is a greater degree of apparent competition predicted using Eq. 4 rather than Eq. 3. In fact, prey 2 is excluded completely for a range of low intrinsic growth rates in the case of Eq. 4. Even when prey 1 is at its carrying capacity, and prey 2 is near zero density, the predator's attack rate on prey 2 is approximately 0.89 according to Eq. 4 with η fitted using a total prey density of 1. Figure 3C shows the consequences of fitting η using the equilibrium density of prey 1 when it is alone with the predator (0.2) for the total prey density. The population sizes shown in Fig. 3C are similar to those in Fig. 3A (based on Eq. 3). The maximum difference between the population size of prey 1 predicted in Fig. 3A and C is equal to approximately 4.6% of the population size given in Fig. 3A, although Fig. 3C predicts less apparent competition between prey than does Fig. 3A. Nevertheless, the message from this figure is that there is little difference between the indirect effects predicted by Eqs. 3 and 4, if the latter is fitted based on a total prey density equal to the ambient density in the system with the predator. Similar results can be obtained using Eq. 5.

Zero handling time is not required for stability. Figure 3D represents the indirect interactions between prey that occur when there is a significant handling time, but the system is stable due to a relatively inefficient predator. The two sets of lines in that graph correspond to Eqs. 3 and 4 for switching, and they are extremely similar. The value of η in Eq. 4 was fitted using a total prey density of 1, which is similar to the one-prey equilibrium density with a predator: $R = 0.8$. In general, for stable systems, values of η based on the equilibrium prey density together with the predator lead to a reasonably close correspondence between densities predicted by Eqs. 3 and 4.

The comparison of population sizes under Eqs. 3 and 5 is not shown explicitly. The best fit of Eq. 5 to Eq. 3 is, of course, when $\delta = 0$, since the two expressions are then identical. Provided that the parameter δ is significantly less than the equilibrium densities of either prey when together with the predator, it has little effect on the equilibrium population sizes. When δ is similar in magnitude, it significantly reduces the extent of switching and can bring about exclusion of the slower-growing prey.

The dynamics of unstable systems

Given that $C_1 z_1 hK > 1$, textbook results (e.g., Case 1999) show that the single prey system with no prey immigration will undergo limit cycles if the equilibrium prey density is sufficiently low. Stability conditions for

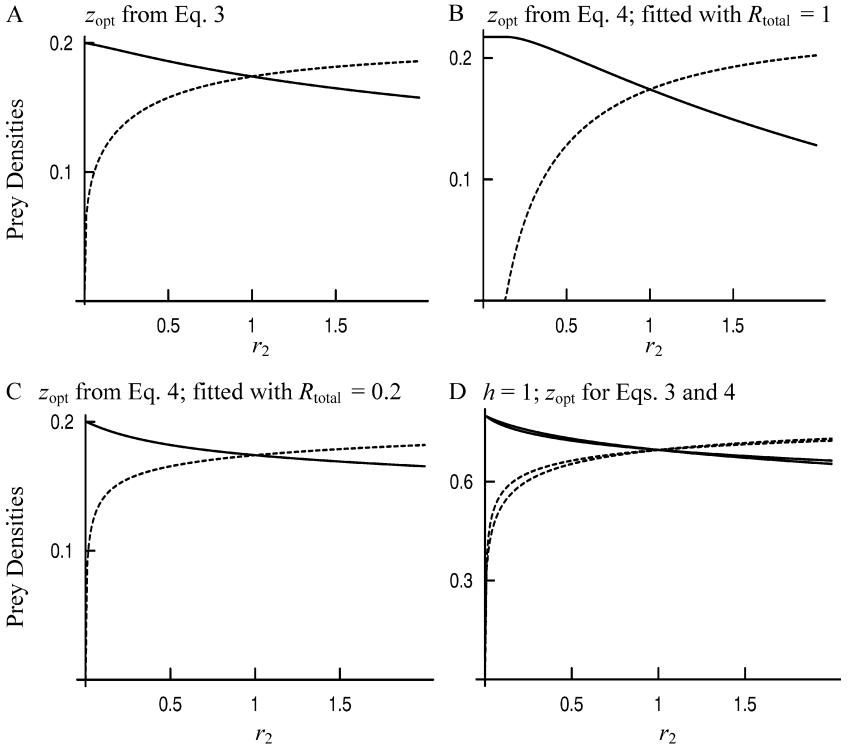


Fig. 3A–D Densities of the two prey (prey 1 *solid line*; prey 2 *dashed line*) as a function of the intrinsic growth rate of prey 2 in stable systems; a comparison of switching Eqs. 3 and 4. In all cases shown, the tradeoff is moderately nonlinear with $\omega = 0.8$. The remaining parameters for **A**, **B**, and **C** are identical to those in the lower panel of Fig. 2, with zero handling time. **A** assumes Eq. 3, **B** assumes Eq. 4 using a total prey density of 1 to fit the parameter η (as in Fig. 1), and **C** assumes Eq. 4 using a total prey density of 0.2 to fit the parameter η. **D** differs in having a nonzero handling time ($h = 1$), and a different predator death rate ($D = 0.08$). The two sets of lines correspond to the predictions made using Eqs. 3 and 4. Equation 4 predicts a lower R_2 and a higher R_1 at $r_2 = 2$ (the right hand side of the graph)

the system with Eqs. 2a–c combined with either Eq. 3 or Eq. 4 are considerably more complicated and must be evaluated numerically. However, it remains true that an efficient predator or large carrying capacities of either prey will generate cycles. In this case, the ultimate dynamics cannot be fully described by a single number. However, the arithmetic mean population sizes are probably the best single measure of the abundance of a cycling species (see Abrams et al. 1998; Abrams 1999). This section will describe interspecific effects in cycling systems, and compare the predictions of the different switching formulas for these cases.

As in the previous section, we do not attempt to describe the differences between stable and unstable systems or between Eqs. 3 and 4 for unstable systems over the entire parameter space entailed by this model. The purpose is simply to show that there can be quite significant differences in both of these comparisons. We again assume a moderately nonlinear tradeoff, with $\omega = 0.8$, and assume that both prey have equal carrying capacities and energetic values (both scaled to 1), and examine a limited range of the other parameters. The best fit of Eq. 4 to Eq. 3 is obtained when $\eta = 10.192$ if total prey density is 0.2, or $\eta = 2.0384$ if it is 1.

Figure 4 shows the dynamics predicted by the two models for a range of r_2 with fixed values of the other parameters. The model based on Eq. 4 has $\eta = 2.0384$. There are obvious differences in dynamics, with greater asynchrony in dynamics and greater differences in mean prey densities in the model based on Eq. 4. This model also predicts exclusion of prey 2 for $r_2 <$ approximately 0.225, while the model based on Eq. 3 predicts coexistence for all values of r_2. Figure 5 shows how average densities of the two prey species change as a function of r_2 based on the two systems shown in Fig. 4, and an additional case with Eq. 4 for z_{opt} fitted using a total prey density of 0.2. Thus, the three panels correspond to the three panels for stable systems shown in Fig. 3A–C. Unlike the stable cases, this unstable system based on Eq. 4 yields densities that often differ significantly from those for Eq. 3 for both values of η. Moreover, there are qualitative differences in the nature of the indirect effect of prey 2 on prey 1 that are predicted by Eqs. 3 and 4 for z_{opt}.

Using Eq. 3 (Fig. 5A) results in indirect effects in cycling systems that superficially appear relatively similar to those for similar stable systems shown in Fig. 3D. However, there are a number of significant differences. There is a large range of initial r_2 values (roughly from 0.5 down to 0.1) where a small or moderate decrease in r_2 actually decreases the mean density of prey 1, rather than increasing it. This effect appears to be due to the decreased amplitude of cycles in prey densities that occur for these cases, because of lower synchrony in the cycles of the two prey species. The results in Fig. 5A also differ from those in Fig. 3D in having higher prey densities and a smaller slope of the lines relating mean prey densities to r_2. However, in these cycling systems, the effect of prey 2 on prey 1 would generally be classified as apparent competition for most (although not all) initial and final values of the intrinsic growth rate, r_2.

Switching based on Eq. 4 (i.e., Fig. 5B, C) often results in between-prey interactions that cannot be classified as apparent competition, and there is little difference in the interactions predicted for different values of η (i.e., Fig. 5B is similar to Fig. 5C). If both prey species have

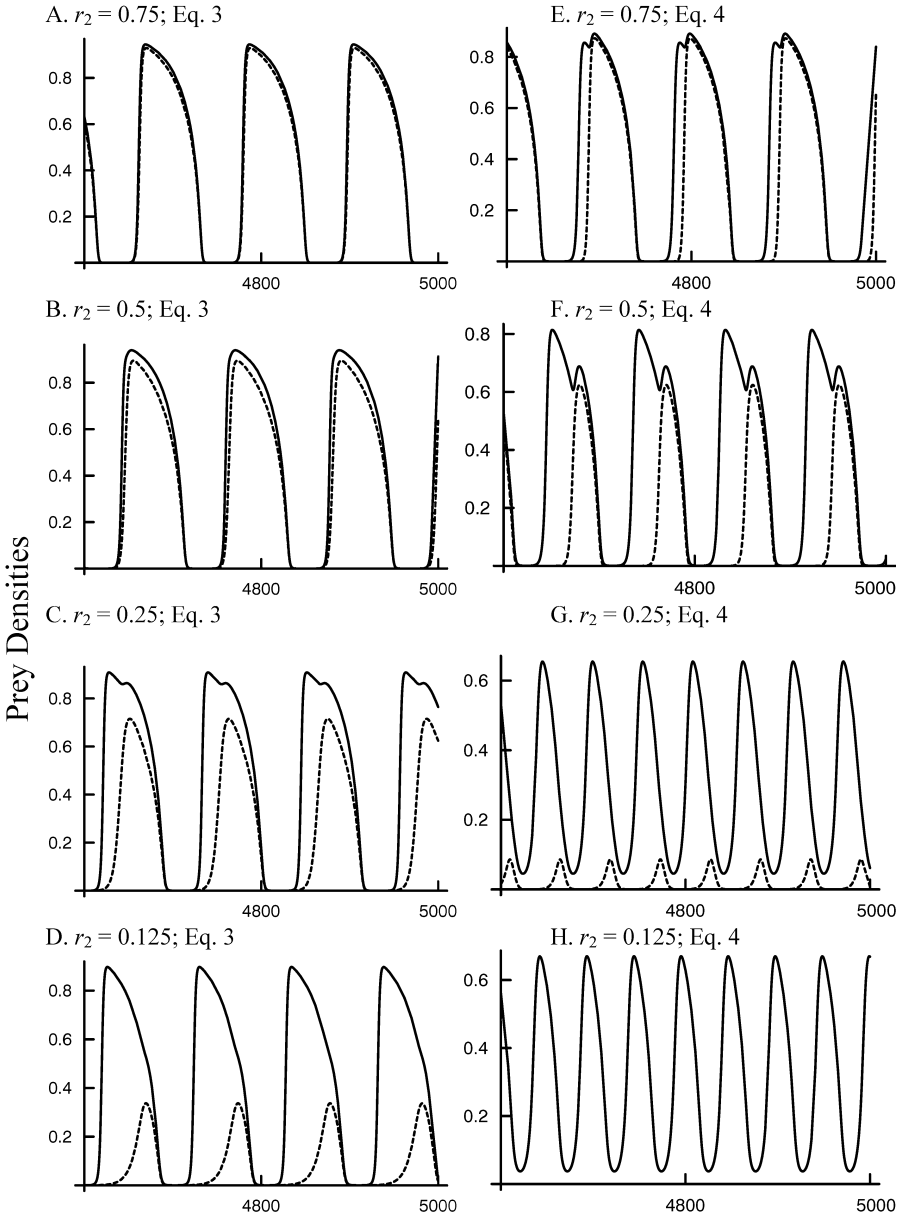


Fig. 4A–H The dynamics produced by Eq. 2 with switching described by either Eq. 3 or Eq. 4 (fitted with a total prey density of 1) for several intrinsic growth rates of prey 2. The other parameters are: $C_j = 5$; $h = 1$; $B = 0.1$; $K_j = 1$; $i_j = 0$; $r_1 = 1$; $D = 0.05$; $e = 1$; and $\omega = 0.8$. The *solid line* is prey 1 and the *dashed line* is prey 2. Each panel assumes a different intrinsic growth rate for species 2. Values with $1 > r_2 > 0.75$ are very similar to those shown for $r_2 = 0.75$. **H** does not show prey 2 because it is excluded at this growth rate

identical intrinsic growth rates of 1, then removing either species will result in a substantial decrease in the population size of the other. When r_2 is initially greater than r_1, then reducing r_2 increases the mean density of prey 1, as would be expected from the usual positive effects of each prey on the shared predator. However, the change in density in this case is much larger (for a given change in r_2) than it would be in a stable system. In all cases where $r_1 \neq r_2$, the difference in densities of the two species is much greater than in a comparable system in which Eq. 3 is used to describe z_{opt}. One reason for this increased indirect effect when Eq. 4 describes switching is the increased asynchrony in the fluctuations of the two prey. When the dynamics of the prey species are synchronized, the predator exhibits little preference, and the splitting of its foraging 'attention' benefits species, offsetting the higher total predator density. Asynchrony means that at least one prey is often abundant when the other is rare, and the former then experiences nearly the full foraging attention of the predator at those times.

Asynchrony arises under Eq. 4 because the predator becomes quite generalized in its diet when both prey are rare, which happens repeatedly during the course of most predator-prey cycles. This means that the slower-growing prey suffers relatively heavy predation when it is rare, and therefore does not recover, following collapse of the predator population, until well after the faster growing prey has recovered. Under Eq. 3, the predator still preys much more heavily on the faster-growing prey when both prey are rare, maintaining approximately equal densities of both. Thus, there is much less of a delay in the recovery of the slower-growing prey species. These features are evident in Fig. 6, which shows both the prey dynamics and the dynamics of the choice trait,

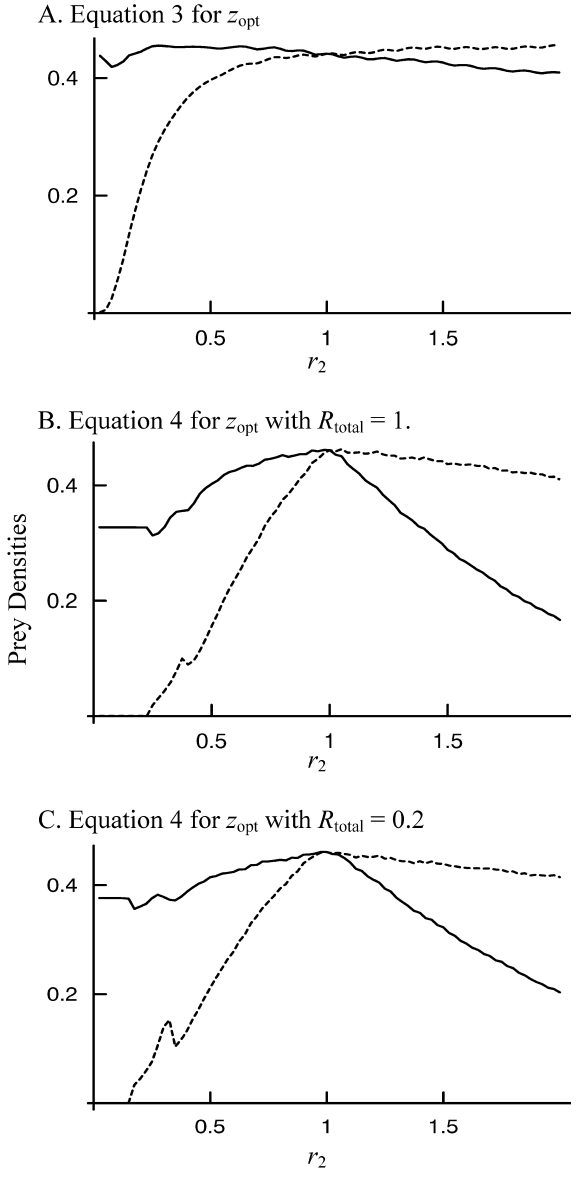


Fig. 5A–C The average densities as a function of the intrinsic growth rate of prey 2 in comparable models based on Eqs. 2 combined with either Eq. 3 (**A**) or Eq. 4 for z_{opt} (**B**, **C**). **B** assumes a total prey density of 1, while **C** assumes a total density of 0.2. The parameter values are identical to those used in Fig. 4. The *dashed line* again denotes prey 2. The only case where all models predict identical densities is when both resources have identical growth functions, resulting in perfectly synchronized cycles with $z_{opt} = 1/2$ in both models

z, under the two models of switching. For the parameters shown in Fig. 6, the two switching formulas produce qualitative different dynamics. Because there is a wide range of prey densities in a cycling system, there is no value of η in Eq. 4 that can provide a good approximation to Eq. 3 over the full range of densities.

Using a different model (i.e., Eq. 5) to model density-dependent switching changes some of the quantitative results, but most of the same qualitative features produced by Eq. 4 are evident. Figure 7 compares the indirect interactions between prey species in an example with switching based on Eq. 3 and an otherwise identical example based on Eq. 5 with $\delta = 0.01$. The difference in indirect effects is qualitatively similar to the difference between Fig. 5A and C, although the average densities differ somewhat.

The examples discussed here show that cycling can greatly alter the indirect effect between prey species via a shared predator that switches. They also show that relatively similar models of switching can predict dramatically different dynamics, particularly in cycling systems. However, we have only presented a small number of examples. There are 12 parameters in Eqs. 2a, b, and c combined with Eq. 3, even after assuming equal profitability of the two prey. Although some of these can be combined by scaling, a complete examination of interactions across all of parameter space is not possible. However, we did examine a number of different asymmetries in the parameters characterizing the two prey species: e, K, and C. None of the cases we examined greatly altered the conclusions that cycling produced significant changes in the indirect effects between prey species, and that the alternative formulas for switching produced significantly different dynamics when systems cycled. When the slower-growing prey was characterized by lower energy content and handling time ($e < 1$), Eq. 3 predicted a larger population size for the slower-growing prey over a wide range of relative growth rates (r_2), while Eq. 4 (or Eq. 5 with a large enough δ) both predicted greater mean densities for the faster growing prey.

We have argued that much of the difference between the densities and dynamics predicted by the two formulas for switching in cycling systems depends on the difference in synchrony of the fluctuations of the two prey species. Asynchrony arises under Eqs. 4 or 5 because of repeated periods when both prey have very low densities. Factors that increase minimum prey densities would therefore be expected to decrease the difference between models where switching does or does not change at low densities. One such factor is immigration rate of prey (i in Eq. 2), which we have assumed to be zero in the preceding examples. Calculations show that significant immigration does have the predicted effect of reducing asynchrony and reducing the difference in mean densities between the prey species. Higher synchrony without immigration can occur in models in which minimum prey densities in cycling systems are greater, due to prey refuges, limited predator hunting at low prey densities, or other factors.

Discussion

Switching by a predator in a stable two-prey system generally reduces the indirect interaction between prey species, making each species less sensitive to changes in either its own or the other prey's intrinsic growth rate.

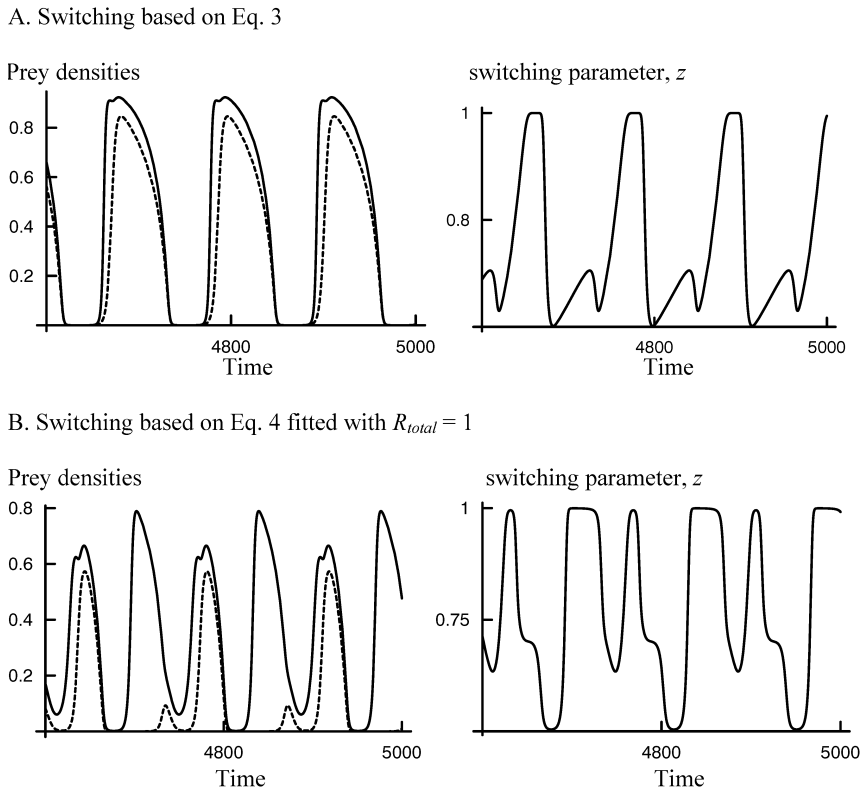


Fig. 6A, B Examples of differences in the dynamics of both prey populations and the switching trait, z, produced by Eqs. 3 (**A**) and 4 (**B**) for a case where the qualitative dynamics differ considerably (parameters as in Figs. 5A, B, with $r_2 = 0.35$. The system using Eq. 4 has predators that become considerably more generalized (z much closer to 1/2) when both prey are rare. The *solid* and *dashed lines* on the left hand panels again correspond to prey 1 and 2, respectively

Under stable conditions, possible changes in switching at prey densities much lower than equilibrium levels have very little effect on the long-term indirect interactions between prey or on conditions for coexistence. However, predator-prey systems are known for their tendency to cycle. Although switching in cycling predator-prey systems has been studied for specific models by Matsuda et al. (1987), Krivan (1997), Abrams (1999), and van Baalen et al. (2001), there has not yet been an analysis of how switching affects the indirect interactions between prey species via their shared predator. The results presented here show that an increase in the intrinsic growth rate of one prey can have a positive effect on the density of the other prey in cycling predator-prey systems. Moreover, switching that becomes less effective at low total prey densities can produce dynamics and indirect effects that differ greatly from purely frequency-dependent switching. In a cycling system, diminished switching at low total densities often leads to relatively high predation pressure on the slower-growing prey species when it is rare, leading to significant asynchrony in prey population cycles, or causing extinction of that prey. As a result, indirect interactions between the two prey species are often much stronger, and coexistence more difficult, than in cases where switching is independent of total density.

Previous work (Abrams 1999, 2000; Abrams and Kawecki 1999) has shown that a time lag in switching can lead to asynchrony between prey in cycling predator-prey systems, and that this asynchrony can alter indirect effects between prey species. The work presented above shows that asynchrony can arise without any behavioral time lag: the present models all assume that behaviors change instantaneously with population densities.

One seldom-noted consequence of switching is that its effects can mask deterioration in the growth rate of a single prey species. In the absence of switching, deterioration in a predator-multiple prey system would be reflected in a pronounced decline in the density of the prey in question. However, as shown by many examples here, the equilibrium density of the prey may be relatively insensitive to its own growth rate, due to increased feeding by the predator on the alternative prey. However, eventually, the prey will decrease very rapidly with further decline in r_2, once that growth parameter is low enough. In the case of cycling systems, the mean density of the prey that initially has a larger r may actually increase with decreases in its own intrinsic growth rate (e.g., Fig. 5B, C), although this trend reverses sharply once its intrinsic growth rate drops below that of the other prey species. For examples of similar phenomena in other systems, see Abrams (2002).

A search of 21 ecological journals for articles on switching (the JSTOR archive; using keywords 'switching' and 'predation' in the abstract) revealed fewer than ten empirical studies of switching over the period 1980–2000. Most of these studies used experiments similar to those carried out by Murdoch in 1969. This scarcity of new investigations of switching cannot be explained by a demonstrated rarity of switching in earlier studies, or by a lack of important ecological

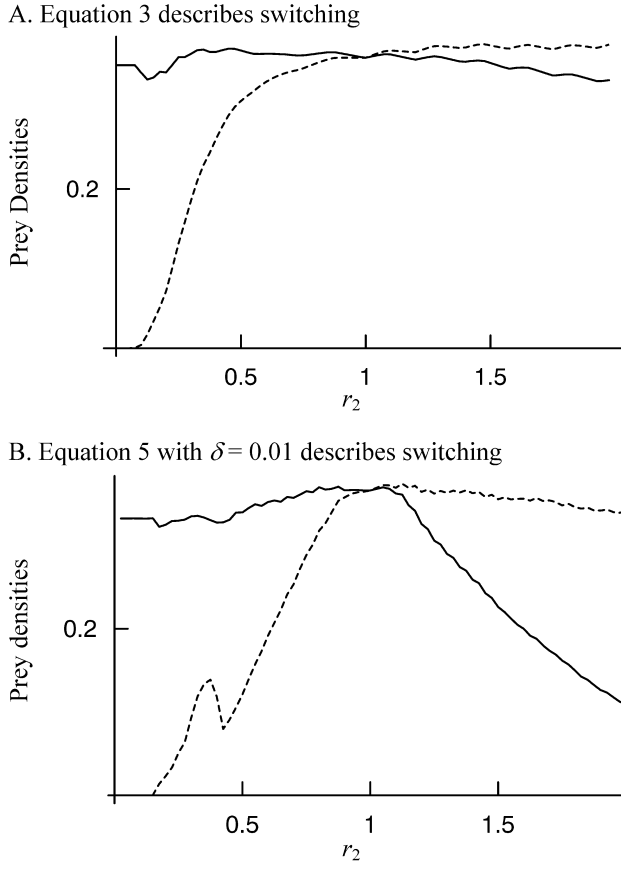


Fig. 7A, B The approximate average densities of the two prey species as a function of r_2 for switching based on Eq. 3 (**A**) and Eq. 5 (**B**). The parameters are: $C_i = 5$; $h = 1$; $B = 0.1$; $K_j = 1$; $i_j = 0$; $r_1 = 1$; $D = 0.04$; $e = 1$; and $\omega = 0.8$. In **B**, which is based on Eq. 5, $\delta = 0.01$

implications of switching (see Introduction). Some experiments have studied long-term population dynamics in systems where switching was likely to occur [see discussion of Bonsall and Hassell (1997) in Abrams and Kawecki (1999)], but the role of switching in producing the final outcome was not explored. We hope that our theoretical work will help convince empirical biologists that there are important and unanswered questions regarding both the dependence of switching on absolute densities and the long-term dynamics produced by switching.

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